# OPERATING SYSTEMS – ASSIGNMENT 1

TASK 2 – Judge the Type of Operating System for Each Environment (3 Marks)
CO1 | Bloom's Level: K5 – Evaluate

**Introduction**
A smart hospital contains devices with different workloads. Patient monitors need predictable response, tablets need mobility and low power use, workstations need interactive computing, and servers need high processing capacity and availability. Therefore, the hospital should select the operating-system type according to the purpose of each environment.

## 1. Comparison of operating-system types

| OS type | Main features | Suitability for the hospital |
|---|---|---|
| **Mainframe OS** | Supports very large numbers of users, transactions and centralised data processing. | Can handle large-scale central transactions, but it is not necessary for every hospital device. A modern server or clustered database is a more suitable practical choice for this design. |
| **Desktop OS** | Provides an interactive graphical interface and supports general applications. | Suitable for doctors' desktop workstations, where staff use clinical applications, view records and interact with the system. |
| **Multiprocessor OS** | Uses two or more processors or CPU cores to execute work in parallel. | Suitable for the MRI imaging server because scan processing can use multiple processors to improve throughput. |
| **Distributed OS / system** | Connects computers at different locations so they can share resources and information. | Suitable for sharing patient records between the main hospital and two partner hospitals. Secure communication and coordination are required. |
| **Clustered system** | Connects multiple independent computers to work together. If one node fails, another can take over. | Best suited to the central patient-record database because fail-over improves service availability and reduces dependence on one machine. |
| **Real-time OS (hard / soft)** | Provides predictable task scheduling. Hard real-time systems must meet strict deadlines; soft real-time systems aim to meet deadlines but occasional misses may be tolerated. | A hard real-time approach is appropriate for bedside alarms because the monitor must raise an alarm within 10 milliseconds. Soft real-time is not sufficient for this strict safety requirement. |
| **Handheld / mobile OS** | Designed for portable devices, touch interfaces, wireless connectivity and power management. | Suitable for nurses' tablets because they are portable and must conserve battery while securely accessing patient information. |

## 2. Recommended OS type for each hospital environment

| Environment | Recommended type | Reason |
|---|---|---|
| Bedside patient monitors | **Hard real-time / embedded OS** | It gives predictable response for the 10 ms alarm deadline and uses limited device resources. |
| Nurses' handheld tablets | **Handheld / mobile OS** | It supports touch interaction, wireless access and battery management. |
| Doctors' workstations | **Desktop OS** | It supports interactive applications, user accounts and clinical software. |
| Central patient-record database | **Clustered server system** | Multiple nodes and fail-over help keep records available if one server fails. |
| MRI imaging server | **Multiprocessor OS** | Parallel processing improves the handling of large MRI image workloads. |
| Partner-hospital link | **Distributed system** | It enables secure sharing and coordination of records across hospital locations. |

## 3. Symmetric vs. asymmetric multiprocessing for the MRI server

**Symmetric multiprocessing (SMP):** In SMP, two or more processors share the same main memory and operating system. Each processor can run tasks, and the OS distributes work among them. This is suitable for the MRI imaging

server because image-processing tasks can run in parallel. It can improve throughput and use the processors efficiently.

**Asymmetric multiprocessing (AMP):** In AMP, one processor may control the system and assign particular tasks to other processors. It can be useful when tasks have fixed roles, but it is less flexible for a changing imaging workload.

**Evaluation:** SMP is the better choice for the MRI server because it supports flexible parallel processing and load sharing. AMP may be selected for a specialised design with fixed processor roles, but it is not the first choice for a general imaging workload.

## 4. Clustered system with fail-over for the database server

The central database should use at least two server nodes connected to shared or replicated storage. The active node handles database requests, while another node monitors its health and is ready to take over. If the active node fails, the standby node performs fail-over and resumes the database service. Data replication, backups and regular fail-over testing are required to reduce data loss and downtime. A cluster improves availability, although it does not remove every risk: network, storage or software failures can still affect the service.

## 5. Alternative concept: one large multiprocessor machine instead of a cluster

If the database used one large multiprocessor server, it could process many requests using several CPUs and may be simpler to manage than multiple cluster nodes. However, the entire machine would be a single point of failure. A hardware, power or operating-system failure could stop database access until the machine is repaired. A cluster provides another node that can take over, potentially within seconds when correctly configured and tested.

**Availability comparison:** A single multiprocessor machine has no automatic machine-level fail-over unless extra redundancy is added. A clustered design improves availability because a standby node can take over when the active node fails. The exact recovery time depends on cluster configuration, failure detection, storage and database recovery; it should be measured rather than assumed.

**Cost comparison:** One large server may have a lower initial purchase and administration cost. A cluster generally costs more because it needs multiple servers, networking, replicated or shared storage and cluster software. However, the extra cost buys better availability and reduces the risk of a long service interruption.

**Throughput and time complexity:** With P processors, a parallel database workload may run faster as work is divided among processors. In an ideal case, the processing time for a parallelisable workload of size N can approach O(N/P), compared with O(N) on one processor. Real performance is lower because of coordination, locking, memory and I/O bottlenecks. Adding processors does not guarantee a proportional speed-up.

**Memory and space complexity:** A single large server can provide a large shared memory space, but it is limited by the maximum RAM supported by that machine. A cluster has memory in each node; the total installed RAM may be larger, but it is not automatically one shared memory space. Distributed database work may need replication and caching, which use additional memory and storage.

## 6. Final recommendation

The hospital should use a hard real-time or embedded OS for bedside monitors, a mobile OS for nurses' tablets, a desktop OS for doctors' workstations, a multiprocessor system for MRI processing, a clustered server for the central database, and a distributed system for sharing records with partner hospitals. For the MRI server, symmetric multiprocessing is preferred. For the database, a cluster with fail-over is safer than relying on one large machine because patient records must remain available.

**Conclusion:** No single OS type is best for every hospital environment. Selecting the appropriate type for each device provides predictable alarm response, efficient processing, mobility, secure record sharing and high database availability.

*Prepared from Task 2 of the uploaded CSA04 Operating Systems Assignment 1. Add your name, register number and any faculty-required diagrams or measured evidence before submission.*